

INSERTION SORT -

The Insertion sort inserts each element in proper place. Suppose an array a with n elements $a[0], a[1], a[2], \dots, a[n]$ is in memory.

The insertion sort algorithm scans a from $a[0]$ to $a[n-1]$, inserting each element $a[k]$ into its proper position in the previously sorted subarray $a[0], a[1], a[2], \dots, a[k-1]$. That is.

Pass 1 $a[0]$ is trivially sorted.

Pass 2 $a[1]$ is inserted either before or after $a[0]$ so that $a[0], a[1]$ is sorted.

Pass 3

Pass N $a[n-1]$ is inserted in its proper place in $a[0], a[1], a[2], \dots, a[n-2]$ so that the array is sorted.

The element inserted in the proper place is compared with the previous elements and placed in between the i th element & $i+1$ th element if

element $\geq i$ th element.

element $\leq (i+1)$ th element

Let us take the elements -

82 42 49 8 92 25 59 52

Pass 1 - 82 42 49 8 92 25 59 52

Pass 2 - 82 42 49 8 92 25 59 52
42 82 49 8 92 25 59 52

elements: 42 82 49 8 92 25 59 52

Pass 3

	42	82	49	8	92	25	59	52
		↑	↓					
	42	49	82	8	92	25	59	52

Pass 4

	42	49	82	8	92	25	59	52
	↑			↓				
	8	42	49	82	92	25	59	52

Pass 5

	8	42	49	82	92	25	59	52
		↑			↓			
	8	25	42	49	82	92	59	52

Pass 6

	8	25	42	49	82	92	59	52
			↑			↓		
	8	25	42	49	59	82	92	52

Pass 7

	8	25	42	49	59	82	92	52
				↑			↓	
	8	25	42	49	52	59	82	92

Pass 8

	8	25	42	49	52	59	82	92
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ALGORITHM - InsertionSort ($A[]$, Item)

Let a be an array of n elements which we want to sort,

1. Set $K=1$
2. for $K=1$ to $(n-1)$

Set $temp = a[K]$

Set $j = K-1$

While $temp < a[j]$ and $(j >= 0)$ perform the following steps.

Set $a[j+1] = a[j]$

[End of loop structure]
3. Exit

c. Function -

void insertion_sort ($int a[]$, $int n$)

{

int $i, K, j, temp$;

for ($K=1$; $K \leq n-1$; $K++$)

{ $temp = a[K]$;

$j = K-1$;

while ($temp < a[j]$ & & $(j >= 0)$)

{

$a[j+1] = a[j]$;

$j = j - 1;$

$\{$
 $a[j+1] = temp;$

$\{$
printf ("Elements of array after sorting are: \n");

for (i

$\{$